# Active sensing explains cortical variability and improves visual information


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## Two debates in visual neuroscience

**Fixational eye movements**

**Feature or bug?**

Are these tiny movements motor noise, or part of how we see fine detail[5,6]?

**Shared cortical variability**

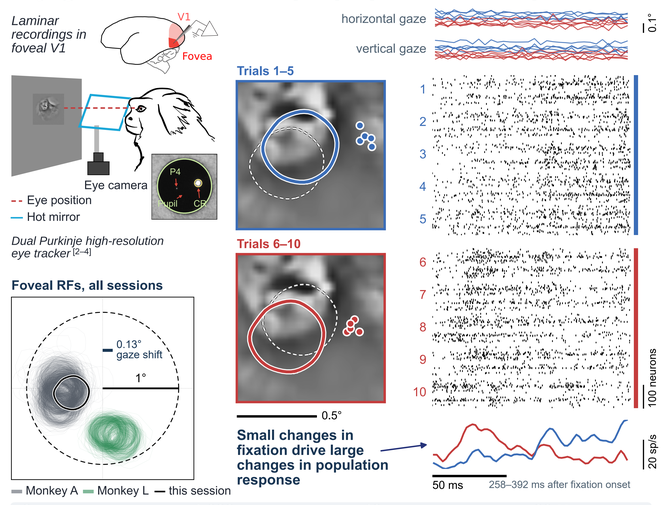
**Noise or signal?**

The same image evokes different responses. Internal noise, or meaningful sensory input?

**How much apparent cortical noise comes from the eye's own movements?**

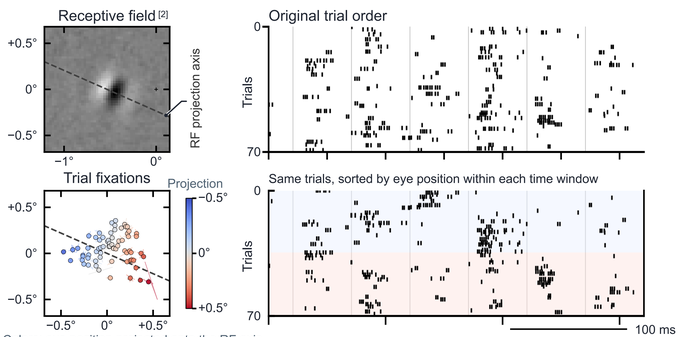
Can that movement-driven activity improve visual information?

## 1 Tiny gaze shifts reshape population responses



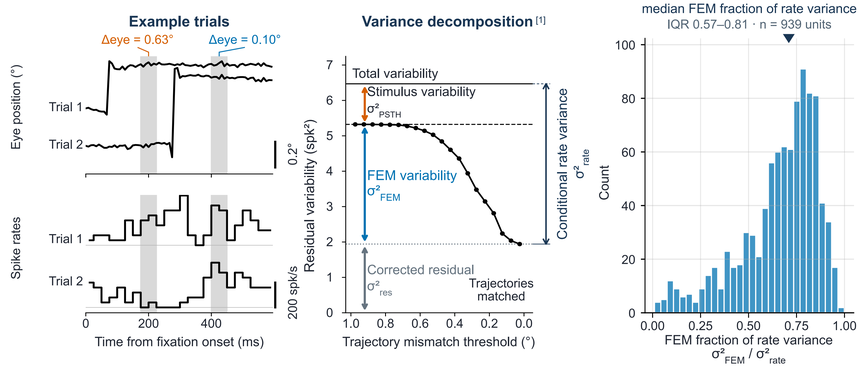

| 2 | 1,022 | 19 | 20 Hz |
|---|---|---|---|
| awake marmosets | foveal V1 units | recording sessions | repeated natural images |

## 2 One neuron's "noise" tracks the eye

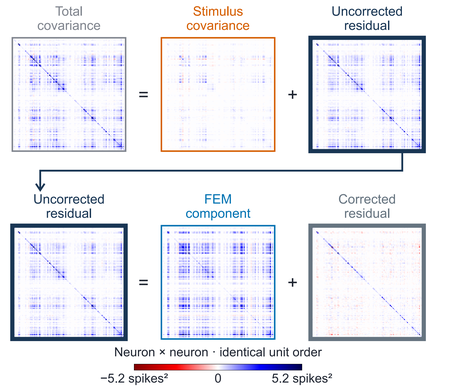


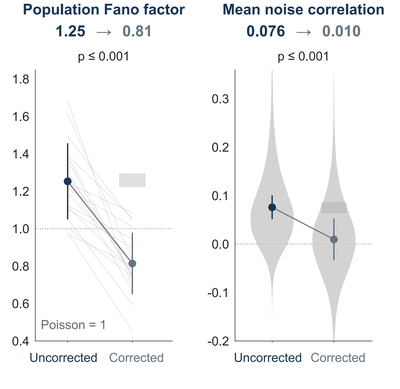

Colors: eye position projected onto the RF axis.

## 3 Eye movements drive most single-cell rate variance (71%)



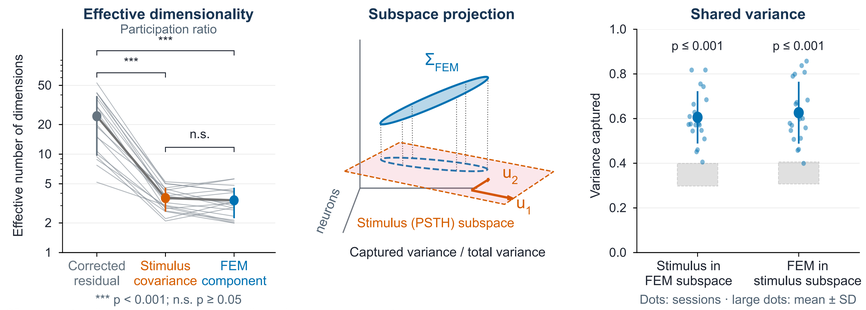

## 4 Correcting for gaze removes shared noise



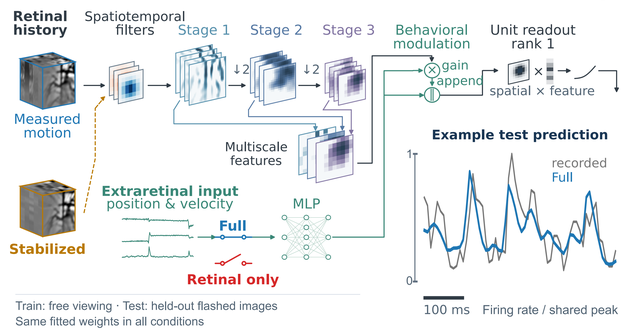

19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

## 5 Eye-driven variability follows stimulus dimensions

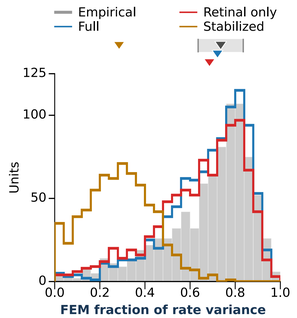



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

## 6 Gaze-dependent retinal input predicts single trials



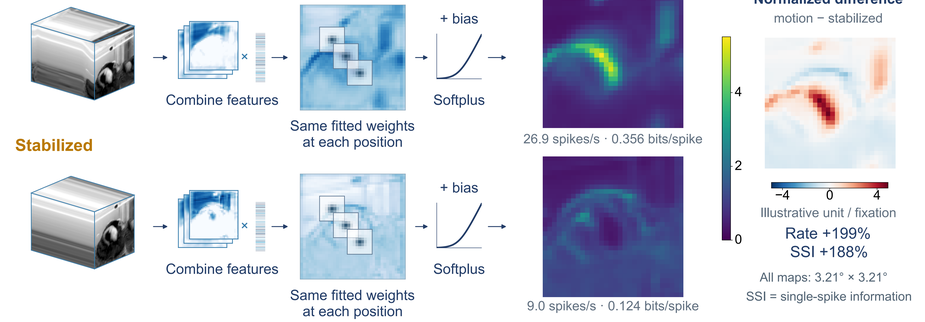

Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions




Triangles: medians; band: empirical ±0.1.

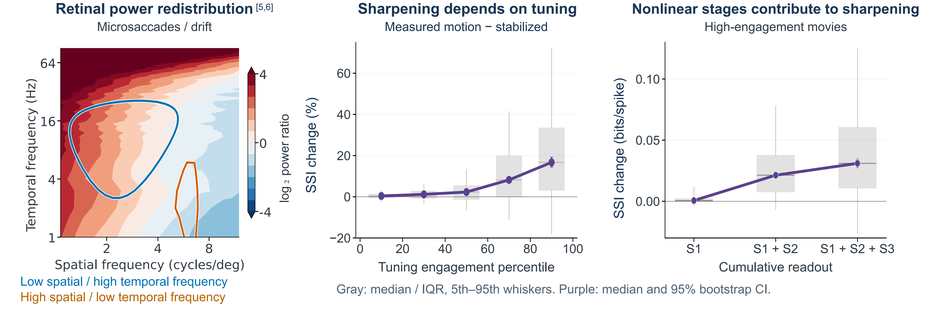
| Median fraction of conditional rate variance explained<br>962 units · 19 sessions | Trial average | Full | Retinal only | Stabilized |
|---|---|---|---|---|
| | 0.139 | 0.235 | 0.226 | 0.042 |

## 7 In the model, eye motion sharpens spatial coding




## 8 Sharpening grows with tuning and model depth

725 units




Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.
Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).


**References**
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012



**Funding**
NEI K99-EY037839-01 (D.P.R.); R01-EY030998 (J.F.M.); R01-EY018363 (M.R.).
Fuster Endowment (A.C.H.); NEI K99/R00-EY032179 (J.L.Y.); NIH/NEI T32-EY007043 (T.K.).



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**Acknowledgements**
Additional preprint coauthors: Dekel Galor · Ruei-Jr Wu · Michele Rucci
We thank Yonatan Abrham (Mitchell Lab) for collecting some of the data used in this report.